PAPER_182: UQFF Complete Variable Reference Table

This paper provides a complete, authoritative reference table for all variables, parameters, and physical constants used in the Unified Quantum Field Framework (UQFF). Twenty-plus variables are catalogued with their symbols, units, canonical numerical values, physical interpretation, and the equation components they appear in. This constitutes the definitive variable dictionary for the UQFF system and serves as the primary reference for all subsequent whitepaper derivations. The table establishes the calibrated constants: κ = 0.0005 day^{-1}, [SSq] = 0.57, H*SCm ≈ 0.99, U*UA ≈ 0.0001, kη = *10*$^{-113}$, βi ≈ 0.603.

Session: 49 — §2.5 Grok Thread 381a8fe7 Extended Audit

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Abstract

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UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Introduction

The UQFF is a multi-component unified field theory built from four gravity components (Ug1–Ug4), a buoyancy field (Ubi), a magnetic string sum (Um), and an aether tensor correction (A_μν). The full field equation is:

```
F_U(r, t, t_n, \theta) = \sum_{i=1}^{4} U_{g,i} + \sum_{i=1}^{4} U_{b,i} + U_m +
A_{\mu\nu}
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Each term depends on a specific set of physical parameters. This paper provides the complete cross-referenced table of all variables.

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2. Universal Physical Constants (Fixed)

Symbol	Name	Value	Units	Source
`c`	Speed of light	`2.998 \times 10^8`	m/s	SI

Symbol	Name	Value	Units	Source
`G`	Gravitational constant	`6.674 \times 10^{-11}`	$m^3/(kg\cdot s^2)$	SI
`\hbar`	Reduced Planck constant	`1.055 \times 10^{-34}`	$J\cdot s$	SI
`k_B`	Boltzmann constant	`1.381 \times 10^{-23}`	J/K	SI
`\Lambda`	Cosmological constant	`1.089 \times 10^{-52}`	m^{-2}	Planck 2018
`B_{\text{crit}}`	Magnetar critical field	`4.4 \times 10^{13}`	T	QED

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3. UQFF Calibrated Constants

Symbol	Name	Value	Units	Equation
`\kappa`	SCm decay rate	`5.0 \times 10^{-4}`	day^{-1}	All Ereact terms
`[SSq]`	Superconductive squeezing factor	`0.57`	dimensionless	SCm coupling
`H_{\text{SCm}}`	SCm Hamiltonian factor	`\approx 0.99`	dimensionless	Ug2, Ubi
`U_{\text{UA}}`	Unified aether factor	`\approx 10^{-4}`	dimensionless	Ubi, A_μν
`k_\eta`	Quantum coupling constant	`10^{-113}`	dimensionless	Deep vacuum
`\beta_i`	Buoyancy coupling coefficient	`0.603`	dimensionless	All Ubi terms
`\eta`	Aether tensor amplitude	`10^{-22}`	s^2/kg	A_μν

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4. Body-Specific Parameters (Per CelestialBody Struct)

Symbol	Name	Typical Value (Sun)	Units	Equation
`M_s`	Stellar/body mass	`1.989 \times 10^{30}`	kg	Ug2, Ug4
`R_s`	Body radius	`6.96 \times 10^8`	m	Ug1, Um
`R_b`	Boundary radius (buoyancy shell)	`1.496 \times 10^{13}`	m	Ug2 (step fn)
`T_s`	Surface temperature	`5778.0`	K	Plasma coupling
`\omega_s`	Rotation rate	`2.5 \times 10^{-6}`	rad/s	Ug3, Um
`B_s`	Average surface magnetic field	`1.0 \times 10^{-4}`	T	Ug1 (μ_s)
`\rho_{\text{SCm}}`	SCm density at body	`1.0 \times 10^{15}`	kg/m^3	Ereact
`Q_{\text{UA}}`	Body UA charge	`1.0 \times 10^{-11}`	C	Ug2

Symbol	Name	Typical Value (Sun)	Units	Equation
`P_{\text{core}}`	Core pressure	`1.0`	Pa (normalized)	Ug3
`P_{\text{SCm}}`	SCm pressure	`1.0`	Pa (normalized)	Um
`\omega_c`	Core angular frequency	`2\pi/(11\,\text{yr})`	rad/s	Ug1, Ug3, Um

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5. Global Physical Parameters

Symbol	Name	Value	Units	Equation
`\Omega_g`	Galactic angular velocity	`7.3 \times 10^{-16}`	rad/s	Ubi (all)
`M_{\text{bh}}`	Black hole mass (SgrA*)	`8.15 \times 10^{36}`	kg	Ubi, Ug4
`d_g`	Galactic distance scale	`2.55 \times 10^{20}`	m	Ubi, Ug4
`v_{\text{SCm}}`	SCm streaming velocity	`0.99c = 2.958\times10^8`	m/s	Ereact
`\rho_A`	Aether density	`1.0 \times 10^{-23}`	kg/m^3	Ereact
`\rho_{\text{sw}}`	Solar wind density	`8.0 \times 10^{-21}`	kg/m^3	Ug2, Ubi
`v_{\text{sw}}`	Solar wind velocity	`5.0 \times 10^5`	m/s	Ug2
`Q_A`	Global aether charge	`1.0 \times 10^{-10}`	C	Ug2
`\rho_v`	Vacuum energy density	`6.0 \times 10^{-27}`	kg/m^3	Ug4
`N_{\text{str}}`	Number of virtual strings	`10^9`	dimensionless	Um
`T_{s}^{00}`	Stress-energy 00 component	`1.27\times10^3 + 1.11\times10^7`	Pa	A_μν

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6. Coupling and Modulation Parameters

Symbol	Name	Value	Units	Equation
`k_1`	Ug1 coupling	`1.5`	dimensionless	Ug1
`k_2`	Ug2 coupling	`1.2`	dimensionless	Ug2
`k_3`	Ug3 coupling	`1.8`	dimensionless	Ug3
`k_4`	Ug4 coupling	`2.0`	dimensionless	Ug4
`\alpha`	Temporal decay rate	`0.001`	s^{-1}	Ug1, Ug4
`\gamma`	String decay rate	`5.0 \times 10^{-5}`	s^{-1}	Um
`\delta_{\text{sw}}`	Solar wind modulation	`0.01`	dimensionless	Ug2

Symbol	Name	Value	Units	Equation
ϵ_{sw}	Wind buoyancy modulation	0.001	dimensionless	Ubi
δ_{def}	Lattice defect amplitude	0.01	dimensionless	Ug1
C_c	Aether concentration	1.0	dimensionless	Ug4
f_{fb}	Feedback correction	0.1	dimensionless	Ug4
$\phi_{\hat{}}$	String orientation factor	1.0	dimensionless	Um

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7. Derived Expressions

Reactor Efficiency

$$E_{\text{react}}(t) = \frac{\rho_{\text{SCm}} \cdot v_{\text{SCm}}^2}{\rho_A} \cdot e^{-\kappa t}$$

At $t = 0$: $E_{\text{react}} \approx \frac{10^{15} \times (0.99c)^2}{10^{-23}} \approx 8.74 \times 10^{45}$ W/m^3

Magnetic Moment

$$\mu_s(t) = \left[B_s + 0.4\sin(\omega_c t) + B_{\text{SCm}}\right] \cdot R_s^3$$

Stellar DPM Moment Sum

$$\sum_j \mu_j = N_{\text{str}} \cdot \bar{\mu}_j \cdot P_{\text{SCm}} \cdot E_{\text{react}}$$

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8. Solar System Validation Values (t = 0)

Body	F_U Component	Value	Units
Sun	U_{g1}	9.26×10^{22}	N (normalized)
Sun	U_{g2}	9.83×10^6	N (normalized)
Sun	U_m	$2.26 \times 10^{16} \cdot (1 - e^{-\gamma t})$	N (normalized)
Earth	$E_{\text{react}}(0)$	$\approx 8.74 \times 10^{33}$	W/m^3
SgrA*	U_{g4}	$\approx 3.55 \times 10^{45}$	N (normalized)

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9. Conclusion

This variable reference constitutes the canonical dictionary for all UQFF equations. The 20+ variables span three categories: universal physical constants (fixed by SI/CODATA), UQFF calibrated constants (fitted to astrophysical observations with $\kappa = 0.0005$ day^{-1}, β*i = 0.603), and body-specific parameters populated by the APIFetch.py pipeline from SIMBAD/NASA/VizieR. All downstream calculations in PAPER*170–195 reference this table.

References

Source: grok*share*381a8f.txt lines 1730–2800 (UQFF derivation and variable section)

Related: PAPER*170–177 (all UQFF component papers), PAPER*187 (canonical MUGESystem catalog)

CP2 Class: `CoAnQiUQFFVariableReferenceCalculator`